Enhanced document retrieval with topic embeddings

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Abstract

Document retrieval systems have experienced a revitalized interest with the advent of retrieval-augmented generation systems. The accuracy of the retrieval mechanism is known to be a bottleneck in the efficiency of these applications. A particular case of subpar retrieval performance is observed in situations where multiple documents from several different but related topics are in the database. We have devised a new vectorization method that takes into account the topic of the document, making retrieval more accurate.

1 Introduction

1.1 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) systems allow us to create chatbots on large text corpora, such as law corpus, textbooks, and software documentation. RAG was introduced by (Lewis et al., 2020). It has seen a sudden rise in popularity due to the availability of improved large language models (LLMs), especially since the release of LLaMA models (Touvron et al., 2023).

RAG works as follows: (1) Text corpus is split into chunks and each chunk is vectorized. (2) User query is vectorized. (3) A similarity search is performed to find the chunk closest to the vectorized query. (4) The retrieved chunk is fed to the LLM along with the user query. (5) LLM uses this input to generate a free-form response.

One of the main bottlenecks in the performance of RAG systems is the retrieval step. The accuracy of the similarity search depends on multiple factors, including the choice of the similarity search algorithm, embedding method, and size of the indexed corpus. Corpus size becomes especially problematic if we are dealing with very similar documents.

1.2 Proposal

In most cases, we are not dealing with raw text data. Instead, the text comes with ample metadata that can be used to boost the performance of the similarity search. Our proposal is to use the available topic information of each chunk (i.e. document) and add it to the retrieval process. We have successfully implemented two such methods in industrial settings, and this paper attempts to generalize and evaluate these methods. The first method relies on creating a new document embedding by combining the original document embedding with the topic embeddings generated from the entire topic. The second method consists of two steps (1) find the topic, and (2) find the document within the topic.

Our contributions in this paper are as follows:

1. Propose two new methods for using topic metadata during the document retrieval process.
2. Release a law dataset in Azerbaijani that can be used to evaluate these and similar methods.
3. Use this and another Turkish dataset to evaluate these methods.
4. Propose a detailed problem statement for the next stages of this work.

The paper is structured as follows: The next section contains a review of relevant literature on these topics. The third section provides a detailed explanation of the proposed methods. The fourth section outlines our experiments, their explanations, and results. The fifth section is dedicated to a discussion of our results and the main shortcomings of our work, as well as suggestions for future research. We conclude with the paper in the sixth section.

2 Previous works

RAG was introduced by (Lewis et al., 2020) in 2020. Their work underpins more advanced RAG systems that we see today. (Liu et al., 2020) uses this idea to create a code summarization tool. Multimodal RAG systems are also seeing increasing

popularity. (Yasunaga et al., 2023) implements a competitive image and caption generation system with RAG architecture.

Vectorization is an integral part of RAG systems. Traditional vectorization techniques include bag-of-words (Qader et al., 2019), which simply uses word frequency and term frequency-inverse document frequency (Qaiser and Ali, 2018), which additionally takes into account how unique the word is to a particular document.

Relatively modern approaches include Word2Vec, GloVe, and fastText. Word2Vec (Mikolov et al., 2013) can use either continuous bag-of-words or continuous skipgram models. GloVe (Pennington et al., 2014) is a log-bilinear regression model that was trained on a word-word cooccurrence matrix. fastText (Bojanowski et al., 2016) is a skip-gram model that treats words as a combination of character n-grams.

The latest family of text vectorization methods relies on transformer-based language models. BERT (Devlin et al., 2019) is a textbook example of this. These models usually use a masked attention mechanism to learn the context of tokens in a text. This information can later be used in various tasks, such as named entity recognition and sentiment analysis. The main advantage of this approach is that embeddings are contextual, i.e. they change depending on their place in the text. This is not the case with the aforementioned methods.

(Nickel and Kiela, 2017) exploits inherent hierarchical structure in the data during the embedding process. Our work is similar but distinct: We use explicit hierarchy, not implicit. As far as we know, no work has attempted this. (Barz and Denzler, 2019) attempts a similar approach to enhance the performance of image classification models on ImageNet.

All of the mentioned works are either language-agnostic, or English-specific. Our dataset is in Azerbaijani, therefore works for the Azerbaijani language are of special interest to us.

Among the open-source embedding models, some claim to have an understanding of the Azerbaijani language. Google has released a multilingual version of the famous BERT model. We are aware of several unpublished attempts to use this model (directly or by fine-tuning) for various NLP tasks in Azerbaijani, all of them with limited success. Due to this, we avoided the model in our work. Another multilingual model that has some understanding of Azerbaijani is mGPT. This is an unofficial version of the GPT-2 model released by (Shliazhko et al., 2023) that was trained in a text corpus consisting of multiple languages. Its text generation capabilities in Azerbaijani are not satisfactory. This is why we did not use it in the generation part of our RAG systems.

3 Methods

We propose two methods that use topic information during document retrieval: topic-enhanced document embeddings, and two-stage document retrieval.

3.1 Topic-enhanced document embeddings

Here, we create topic embeddings and then use these embeddings to update the original document embeddings in one of two ways.

Step 1: Create document embeddings.

This is part of the traditional RAG pipeline. We simply chunk the document and embed these chunks separately.

Step 2: Separate documents into topics.

We assume the topic information is provided either implicitly or explicitly. In any case, we need to have explicit topic labels by the end of this stage.

Step 3: Creating topic embeddings.

Topic embeddings are supposed to be a vector of the same size as document embeddings. If we are using a neural network like BERT to embed the text, we usually cannot feed the entire topic into the model. We can bypass this by taking an element-wise average of all document embeddings for that topic. If we are using a statistical method like TF-IDF, we can run it on the entire text regardless of the text length. However, we expect both the original document embeddings and the topic embeddings to use the same embedding method.

Step 4: Update the document embeddings

We propose two alternate versions here:

1. *Average method*: Take an average of document embeddings and topic embeddings.

2. *Append method*: Concatenate document embeddings and topic embeddings.

The second method creates a new problem because now the embedding dimension of a query does not match the dimensions of our embedded documents. To solve this problem, we can duplicate the length of our query embeddings.

Table 1: Azerbaijani laws in our dataset.

Law
Criminal Code
Code of Criminal Procedure
Customs Code
Constitution
Forest Code
Civil Procedure Code
Civil Code
Migration Code
Water Code
Land Code
Tax Code
Code on Administrative Violations
Labor Code
Education Law

3.2 Two-stage document retrieval

This is a much simpler approach. We create topic embeddings just like the first method, but we perform retrieval in two stages:

1. Retrieve a topic based on the topic-embeddings.
2. Retrieve a document within that topic.

4 Experiments

To evaluate these approaches, we have curated a dataset from Azerbaijani laws. We have selected 14 major laws and split them into chunks of 2000 characters. You can find these laws in Table 1. OpenAI's "text-embedding-3-small" embedding model has been used to embed the chunks. Each law is treated as a distinct topic. We then used both the average method and the append method to update the document embeddings.

By treating topics (i.e., laws) as cluster labels, we have evaluated the "clustering results" of three methods:

- original embeddings
- averaged embeddings
- appended embeddings

We have used the tSNE algorithm to reduce the number of dimensions to 2 (Cai and Ma, 2022). You can see the results in Figure 1a and 1b.

We have calculated the performance of these "clustering models." They are available in Table 2. As you can see, adding topic embeddings to the original embeddings results in better separation of different topics. The average method performs better than the append method, although we suspect that it may be data-specific. We will perform these experiments on larger and more variable datasets to assess the methods' performance against each other.

Table 2: Performance of original and topic-based clustering. DBI: Davies–Bouldin index, CHI: Calinski–Harabasz index.

Metric	Original	Average	Append
Silhouette	0.01	**0.11**	0.06
DBI	4.60	**2.30**	3.25
CHI	63.42	**253.67**	126.84

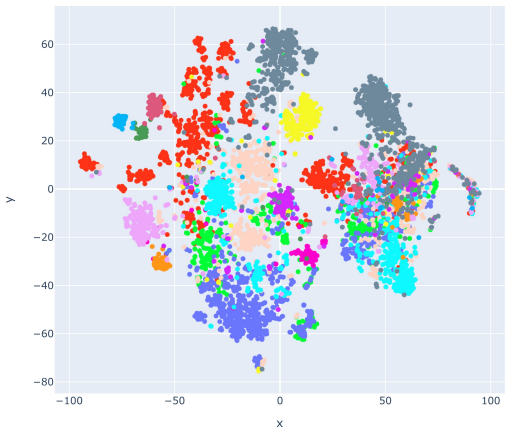
5 Challenges and limitations

The main challenge we are facing is the lack of an end-to-end evaluation dataset. This dataset needs to have the following features:

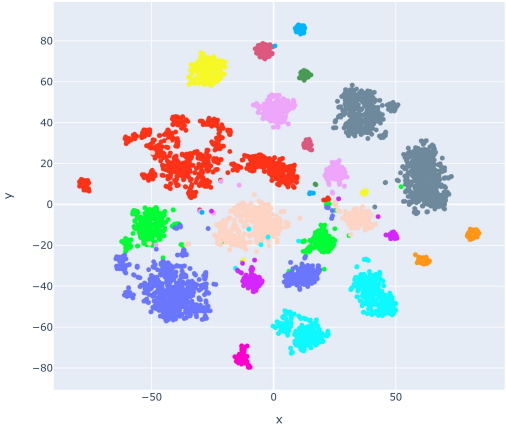
- A text corpus with topic labels
- Natural queries that can be answered based on a part of this corpus.

This first feature is easy to attain. We have already performed some analysis on such datasets. We have attempted to gain the second feature by synthetic query generation, but it was not successful, because generated queries are highly specific to the chunk provided. For example, we used the following query:

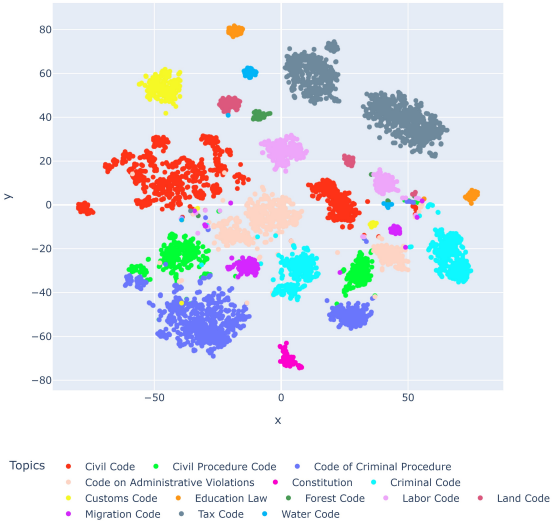
> Given the following context, generate a question that would be asked by a curious citizen:
>
> Article 25. Right to equality
>
> I. Everyone shall be equal before the law and the courts.
>
> II. Men and women possess equal rights and freedoms.
>
> III. The State shall guarantee the equality of rights and freedoms to everyone, irrespective of race, ethnicity, religion, language, sex, origin, property status, occupation, beliefs or affiliation with political parties, trade union organisations or other public associations. Restrictions of rights and freedoms on the grounds



(a)



(b)



(c)

Figure 1: 2D visualization of topics with (a) original embeddings, (b) averaged embeddings, (c) appended embeddings.

of race, ethnicity, religion, language, sex, origin, beliefs, or political or social affiliation are prohibited.

IV. No one may be harmed, granted advantages or privileges, or refused to be granted advantages and privileges on the grounds laid down in Paragraph III of the present Article.

V. Everyone shall be guaranteed equal rights in any proceeding before state authorities and bearers of public authority that decide upon his/her rights and duties.

VI. Persons with impaired health are entitled to all rights and carry all duties enshrined in this Constitution, except in cases when enjoyment of rights and performance of duties is impeded by their limited abilities.

This resulted in the following response:

"What measures are in place to ensure enforcement of Article 25, particularly regarding equality before the law and in courts, as well as the guarantee of equal rights and freedoms regardless of various personal characteristics?"

As you can see, if we generate a large dataset automatically using these prompts, we will end up with questions that exactly match the provided context (i.e., chunk).

Our attempts here were on the translated English version of our dataset. As far as we are concerned, no LLM can generate natural text in Azerbaijani at a level that would be sufficient for this task. Due to these reasons, we are researching the possibility of creating a natural dataset. We can use logs of one of our chatbots in production for this, or we can create an annotator team to build the dataset.

The main inherent limitation of our approach is that it depends on the existence of topic metadata. It would be interesting to devise a method to infer the topic information from the raw data itself, although we do not believe that there is a generalizable approach to this problem.

6 Conclusion

We propose two new methods that utilize the provided topic information in the text corpus to enhance document retrieval. ...

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